PST-DUNet: A Progressive Spectral Topology-Aware Deep Unfolding Network for Hyperspectral Image Super-Resolution

Zhenyu Li, Xiangrong Zhang, *Senior Member, IEEE*, Guanchun Wang, *Member, IEEE*, Huiyu Zhou, Xiao Han, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—Hyperspectral images (HSIs) contain rich spectral fingerprints. However, restricted by hardware bottlenecks, the acquired images often fail to satisfy the requirements for fine-grained observation. Although HSI super-resolution has achieved significant progress, many existing methods still primarily emphasize spatial detail recovery and exploit cross-band information through implicit feature interactions or output-level spectral constraints. Such a spatial-first reconstruction paradigm may provide insufficient cross-band coordination during progressive detail generation, leading to inconsistent interband response patterns and spectral distortion. To address this issue, we propose a progressive spectral topology-aware deep unfolding network (PST-DUNet). In particular, we adopt a progressive low-resolution (LR)–mid resolution (MR)–high-resolution (HR) reconstruction strategy to reduce the difficulty of direct large-scale reconstruction. In the MR stage, a parameterized observation-consistency correction is introduced to constrain the intermediate estimate with the LR observation. Meanwhile, we design an instance-adaptive dynamic spectral topology inference (DSTI) module to infer reconstruction-state-conditioned interband dependencies. Furthermore, a spectral topology rectifier (STR) is introduced to transform the topology-aggregated spectral representation into channelwise gating coefficients, thereby modulating spatial feature updates and alleviating spectral distortion. Extensive experiments on the evaluated public HSI datasets like Pavia Center, Chikusei, and CAVE demonstrate that PST-DUNet achieves superior reconstruction performance under the nonblind degradation setting, providing a favorable balance between spatial detail recovery and spectral consistency preservation.**

***Index Terms*—Deep unfolding network, dynamic graph learning, hyperspectral image super-resolution (HSI-SR), spectral topology.**

I. Introduction

HYPERSPECTRAL imaging acquires spectral radiance information over hundreds of contiguous narrow bands, providing each pixel with a high-dimensional and fine-grained spectral representation. Compared with RGB images or multispectral images that contain only a few broad bands, the spectral "fingerprints" carried by hyperspectral image (HSI) characterize material differences and spectral responses more comprehensively [1], [2], and have therefore become invaluable for a wide range of remote-sensing applications, such as fine-grained land-cover mapping and classification, environmental and agricultural monitoring, and mineral exploration [3], [4], [5], [6], [7], [8], [9], [10]. However, constrained by the sensor's physical aperture, dwell time, and radiometric energy budget, hyperspectral imaging systems inevitably have to reduce the spatial sampling rate. This spatial degradation undermines the effectiveness and reliability of hyperspectral data in remote-sensing analysis [11]. Compared with high-cost hardware upgrades, HSI super-resolution (HSI-SR) offers a practical alternative to enhance spatial resolution and recover fine details under existing observation conditions, and has thus emerged as an important means to improve the usability and application value of HSI.

Although the success of super-resolution for natural images [12], [13], [14], [15], [16], [17] has provided important paradigms for HSI-SR, the objective of HSI-SR goes far beyond producing visually sharper images [18], [19], [20], [21]. From an inverse-problem perspective, HSI-SR is a severely illposed restoration task under substantial loss of spatial information, where spatial high-frequency details are irreversibly discarded during downsampling, and any attempt to recover them inevitably relies on prior assumptions. Unlike natural images such as RGB, the discriminative power of HSI lies in the relative response patterns across bands, including the shape of the spectral signature and the relative variations among bands. When reconstruction introduces band-dependent, inconsistent detail enhancement or intensity shifts, these relative relationships are disrupted, leading to spectral signature distortion at the same spatial location, mismatched interband correlation structures, and increased spectral angle errors, which ultimately compromise target discrimination and the reliability of physical interpretation.

In recent years, deep learning-based HSI-SR methods have made substantial progress. Most existing approaches largely follow the dominant paradigm in natural-image SR, emphasizing spatial-domain texture completion and detail recovery. Meanwhile, HSI-oriented methods have introduced 3-D convolutions [22], spectral attention mechanisms [23],

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The authors are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: xrzhang@mail.xidian.edu.cn).

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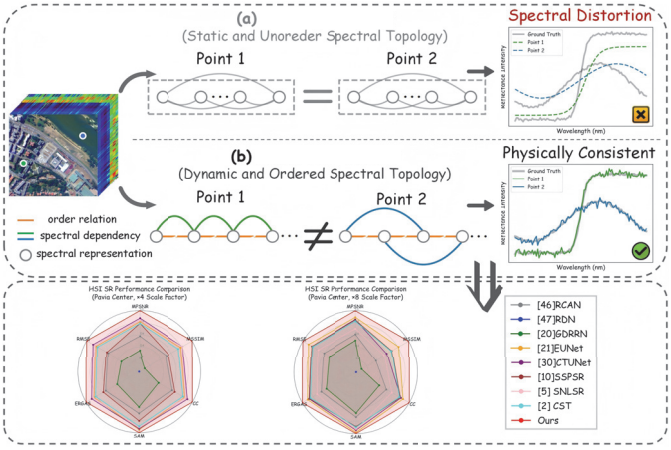


Fig. 1. Motivation of the proposed reconstruction-state-conditioned interband dependency modeling. (a) Existing methods often exploit cross-band information through implicit feature interactions or static feature recalibration. Such strategies may provide insufficient adaptive coordination among spectral bands during spatial detail recovery, resulting in inconsistent interband responses. (b) On the contrary, PST-DUNet represents spectral bands as band nodes and dynamically infers input-dependent interband dependencies for the current reconstruction state. The inferred dependencies are used to guide channelwise spatial feature modulation, thereby improving the coordination between spatial detail recovery and spectral consistency preservation.

and spectral losses such as the spectral angle mapper (SAM) [24] to enhance spatial–spectral representation and spectral consistency. These designs have effectively improved HSI-SR performance. However, in many methods, cross-band relationships are still mainly exploited through implicit feature transformations, feature-level reweighting, or output-level loss constraints. As a result, the interband response relationships are not explicitly modeled during the reconstruction update, which may limit the coordination of spatial detail generation across spectral bands.

These observations point to a more fundamental question in HSI-SR: under severe spatial degradation, how can spatial details be recovered without destroying the relative response patterns among spectral bands? As shown in Fig. 1, without sufficient cross-band coordination, a model may sharpen spatial textures while amplifying interband inconsistencies, resulting in distorted spectral signatures and increased spectral angular errors. Therefore, robust HSI-SR requires moving beyond independent pixelwise mapping and introducing adaptive interband dependency modeling into the reconstruction process. Since the useful dependencies among spectral bands may vary with scenes, materials, and reconstruction states, they should be inferred dynamically rather than fixed as a predefined static graph. This motivates us to use dynamic interband dependency as a structural condition to guide spatial detail recovery and stabilize spectral consistency.

To more effectively mitigate spectral drift during spatial detail recovery and jointly improve spatial fidelity and spectral consistency, we propose a progressive spectral topology-aware deep unfolding network (PST-DUNet). The main idea of PST-DUNet is to combine observation-consistency correction with reconstruction-state-conditioned interband dependency modeling in a progressive reconstruction framework. In particular, PST-DUNet first performs spectral topology-guided feature calibration in the low-resolution (LR) space to obtain a stable initial representation. It then conducts finite-step learned unfolding in the midresolution (MR) space, where each unfolding step consists of an observation-consistency correction and a learned prior refinement. The observation-consistency correction constrains the MR estimate using the LR observation, while the prior refinement is conditioned on the interband dependencies inferred by dynamic spectral topology inference (DSTI). Finally, the spectral topology rectifier (STR) transforms the topology-aggregated spectral representation into channelwise gating coefficients to modulate spatial feature updates. In this way, the progressive recovery of spatial details is coordinated with the dynamically inferred interband dependency structure.

In summary, the main contributions of this work can be summarized as follows.

1) We propose PST-DUNet for HSI-SR, which follows a progressive LR–MR–high-resolution (HR) reconstruction strategy. By performing finite-step learned unfolding in the MR space and introducing observation-consistency correction with the LR input, the proposed framework reduces the difficulty of direct large-scale reconstruction and provides a stable reconstruction path.
2) We develop DSTI to infer reconstruction-state-conditioned interband dependencies from the current image-domain estimate. The inferred dynamic adjacency matrix provides adaptive cross-band relational information for the subsequent reconstruction update, instead of relying on a predefined static spectral graph.
3) We introduce STR to inject the topology-aggregated spectral representation into spatial feature refinement. By generating channelwise gating coefficients from the inferred interband dependency representation, STR improves the coordination between spatial detail recovery and spectral consistency preservation.

The remainder of this article is organized as follows. Section II reviews the related work on HSI-SR and deep unfolding networks. Section III elaborates on the proposed PST-DUNet, detailing the progressive framework, the STR, and the DSTI. Section IV presents the experimental settings, comprehensive comparisons with state-of-the-art methods, and ablation studies to validate the effectiveness of our design. Finally, Section V concludes this article.

II. Related Work

A. Traditional Model-Based Methods for HSI-SR

Early studies on HSI-SR typically start from the imaging degradation process, formulating reconstruction as a constrained optimization problem and restricting the solution space with hand-crafted priors. Given the strong correlation and substantial redundancy along the spectral dimension, low-rank assumptions have been extensively adopted to characterize global spectral structures and improve the well-posedness of estimation. Representative works [25], [26] combine low-rank tensor decomposition with regularizers such as total variation (TV), enforcing global spectral consistency while promoting local spatial smoothness. Beyond low-rank

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modeling, spectral subspace approaches have also been widely used for reduced-dimensional reconstruction. Akgun et al. [27] represent high-dimensional spectral observations as linear combinations of a small number of basis components, enabling reconstruction in a low-dimensional spectral subspace and thereby reducing estimation difficulty and enhancing stability. Sparse representation and dictionary learning constitute another important line of research, where learned dictionaries and sparse codes are leveraged to strengthen high-frequency representation and detail recovery [28]. In addition, Bayesian formulations under a maximum a posteriori (MAP) framework [29], [30] derive objective functions by explicitly incorporating noise models and prior distributions, providing a principled basis for interpretability.

Overall, these methods are advantageous in that they offer relatively strong interpretability. However, they typically rely on static, predefined priors, which limit their ability to adapt to complex, scene-dependent interband relationships.

B. Deep Learning-Based Methods for HSI-SR

Single-image HSI-SR refers to reconstructing an HR-HSI solely from an LR HSI, without relying on any additional HR data. In recent years, deep neural networks have driven substantial progress in HSI-SR thanks to their strong nonlinear modeling capability, and existing approaches can be broadly categorized into three groups.

1) Spatial–Spectral Prior-Learning-Based: Many early works adopt CNNs as the backbone and treat the spectral dimension as the channel dimension. To strengthen spatial–spectral coupling, Mei et al. [31] proposed 3DFCNN, which employs 3-D convolutions to learn joint features across spatial and spectral dimensions. Subsequent studies explored various hybrid convolution designs to further enhance spatial–spectral representations, e.g., using 1-D and 2-D convolutions to model spectral and spatial information separately [32], or inserting 3-D modules into a 2-D backbone to complement spectral feature learning [22], [24], [33]. In addition, strategies such as parameter sharing, group convolution, and recursive residual learning have been used to improve feature reuse and training stability. Li et al. [34] leverage parameter sharing to reduce redundancy. Jiang et al. [35] emphasize joint learning of spatial–spectral priors, strengthening correlations among neighboring bands via manual band grouping and parameter sharing, together with a progressive reconstruction strategy for improved stability. In addition, some works explore transfer learning and explicit spectral constraints. Yuan et al. [36] transfer the LR–HR mapping learned from natural-image SR to HSI-SR and introduce endmember-consistency constraints to alleviate cross-band deviations. Hu et al. [37] exploit spectral priors by transforming reconstruction into a more structurally constrained abundance domain to learn spectral structures more directly.

2) Long-Range Dependency Enhancement-Based: To overcome the limited receptive field of convolutions for modeling long-range dependencies, some studies introduce global relation modeling mechanisms to better exploit nonlocal spatial similarity and cross-range contextual information. Liu et al. [38] adopt a dual-branch architecture consisting of a Transformer branch and a 3-D CNN branch to capture global context and extract local spatial–spectral details, respectively, and enhance spatial–spectral consistency during reconstruction via bidirectional interactions between the two branches. Chen et al. [39] further perform spatial–spectral information aggregation from a cross-scale interaction perspective. On the spatial side, they introduce interactions between local windows and aggregated global features within window-based attention to complement long-range spatial dependencies with linear complexity. On the spectral side, they establish global spectral correlations via interactions between representative spatial–spectral features and global features, avoiding dense attention over the full spectral dimension. In addition, Xu et al. [40] introduce state space models into HSI-SR, using dual spatial–spectral pathways to model long-range dependencies and providing an alternative route for leveraging global context.

3) Deep Unfolding-Based: Another way revisits HSI-SR from a MAP perspective and unrolls the iterative solver into a trainable network, thereby explicitly incorporating degradation consistency and data-consistency (DC) updates during reconstruction. Liu et al. [41] formulates single-image HSI-SR as MAP inference and unfolds the iterative updates into a multistage network, alternating between DC steps and learnable prior updates within each stage. Wang et al. [42] further implement the iteration as a multistage network composed of fidelity-update layers and prior-learning layers, and introduce cross-stage recurrent connections to exploit stage-to-stage recursive information. Tang et al. [43] adopt a similar unrolled structure and enhance the continuity of priors across stages via cross-stage feature fusion.

Overall, these methods have substantially improved spatial detail recovery through the above three strategies, and have also enhanced spectral consistency to some extent. However, interband dependencies are still predominantly handled through implicit feature interactions or loss constraints imposed at the output level, making it difficult to provide a controllable and persistent cross-band coordination mechanism during progressive detail generation and iterative refinement. Consequently, under severe degradations, spectral drift and degraded spectral reliability may still occur. This motivates explicitly modeling input-adaptive interband dependency structures and coupling them with the reconstruction update path to stabilize cross-band consistency and improve reconstruction robustness.

III. Proposed Approach

A. Problem Formulation

Let the HR-HSI be denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H and W represent the height and width in the spatial dimension, respectively, and B denotes the number of spectral bands. The corresponding LR observation is denoted as $\mathbf{Y} \in \mathbb{R}^{h\times w\times B}$, where $h < H$ and $w < W$.

In the process of remote-sensing hyperspectral imaging, spatial resolution degradation is mainly caused by optical

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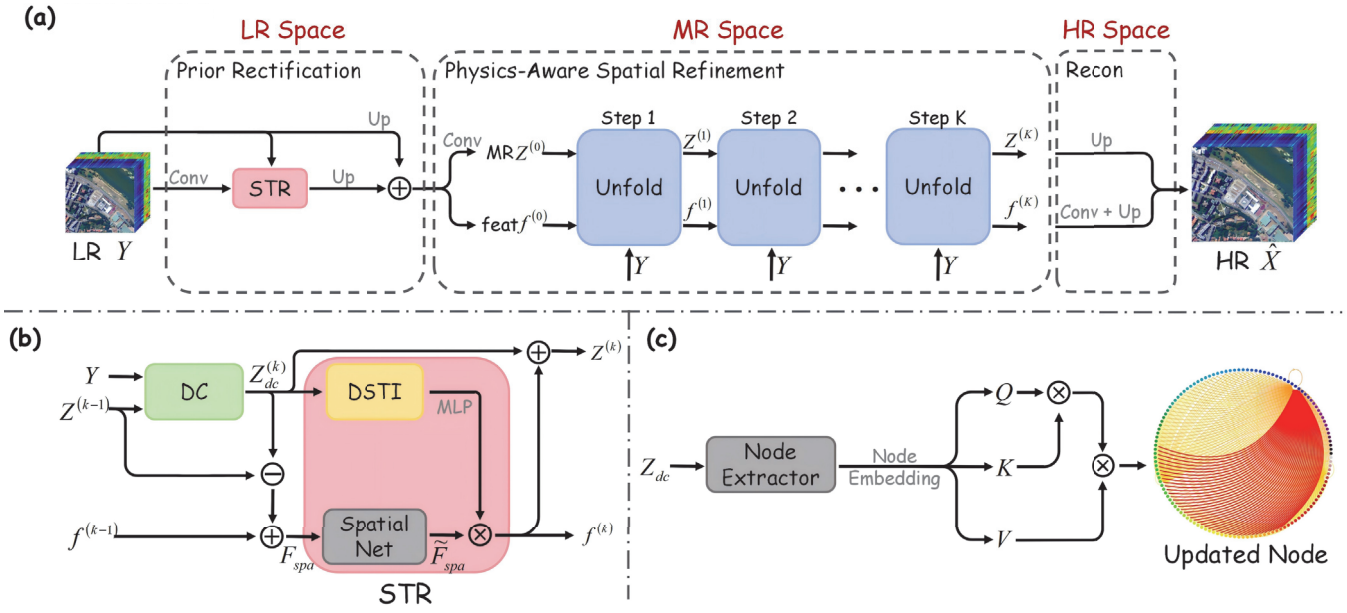


Fig. 2. Framework of the proposed PST-DUNet. (a) Overall architecture follows a fixed LR–MR–HR reconstruction path, including LR feature calibration, MR unfolding refinement, and final HR reconstruction. (b) Detailed structure of the kth MR unfolding step. It contains an observation-consistency correction module that uses the LR observation $\mathbf{Y}$ to correct the current MR estimate, and a learned prior refinement module where STR modulates spatial features according to the interband dependency representation inferred by DSTI. (c) Illustration of DSTI, which represents spectral bands as nodes and dynamically infers a reconstruction-state-conditioned interband dependency matrix.

blurring and spatial downsampling. A commonly used degradation model for HSI-SR can be written as

$$\mathbf{Y} = D_s B_{\text{blur}} \mathbf{X} + N \tag{1}$$

where B_{blur} denotes the spatial blur operator, D_s denotes the spatial downsampling operator, and N represents additive noise. Here, B_{blur} is used to denote the blur operator to avoid confusion with the number of spectral bands B. The degradation operator $D_s B_{\text{blur}}$ maps the HR-HSI to the LR observation along the spatial dimensions, while the spectral dimension is preserved.

Under this degradation model, HSI-SR is usually regarded as an illposed inverse problem. Since spatial downsampling inevitably removes high-frequency spatial information, multiple HR-HSIs may correspond to the same or very similar LR observations. Therefore, recovering X from Y requires not only consistency with the LR observation but also effective spatial–spectral priors to regularize the solution space. The following method is designed to introduce observation consistency and learned interband dependency priors into the reconstruction process.

B. Overview of PST-DUNet

We propose a PST-DUNet, whose overall architecture is illustrated in Fig. 2. PST-DUNet follows a fixed LR–MR–HR reconstruction path. Given the observed LR HSI $\mathbf{Y} \in \mathbb{R}^{h\times w\times B}$, the network first constructs an intermediate MR estimate $\mathbf{Z} \in \mathbb{R}^{2h\times 2w\times B}$ and then reconstructs the final HR-HSI $\hat{\mathbf{X}} \in \mathbb{R}^{rh\times rw\times B}$, where r denotes the target scale factor.

The main idea of PST-DUNet is to combine LR observation consistency with reconstruction-state-conditioned interband dependency modeling. The MR space serves as an intermediate reconstruction domain between the information-limited LR observation and the high-dimensional HR target. Compared with direct LR-to-HR mapping, MR reconstruction reduces the difficulty of a single large-scale upsampling process. Compared with directly performing iterative refinement in the HR space, MR refinement involves a smaller spatial search space and can still be connected to the LR observation through MR-to-LR consistency correction. Based on this design, PST-DUNet consists of three functional phases.

1) Spectral Topology-Guided Feature Calibration in LR Space: The first phase operates in the LR space. Since the LR observation is the direct input measurement, it provides reliable spectral response information for initial feature construction. The network first extracts shallow LR features from $\mathbf{Y}$. Then, DSTI and STR are used to infer interband dependencies from the LR observation and to generate channelwise modulation for the LR spatial features. Therefore, this phase provides a spectrally coordinated initial representation for the subsequent MR reconstruction.

2) Observation-Consistency-Guided Refinement in MR Space: The second phase performs the main reconstruction refinement in the MR space. Starting from the initial MR estimate, PST-DUNet conducts a finite number of learned unfolding steps. Each step contains two parts: an observation-consistency correction that uses the LR observation to adjust the current MR estimate, and a learned prior refinement that is conditioned on the interband dependencies inferred by DSTI and injected through STR. The detailed formulation of the MR unfolding step is given in Section III-C.

3) Final Reconstruction in HR Space: After K MR unfolding steps, the refined MR feature representation $f^{(K)}$ is mapped

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to the HR space by a learnable subpixel upsampling module. In particular, the MR-to-HR transition adopts a PixelShuffle-based upsampling operator with scale factor $r/2$, followed by convolutional reconstruction. It is combined with the upsampled MR estimate $\mathbf{Z}^{(\mathbf{K})}$ to obtain the final HR reconstruction $\hat{\mathbf{X}}$. The HR reconstruction is the only output used for supervised training.

It should be noted that the LR, MR, and HR phases are jointly optimized in an end-to-end manner. PST-DUNet does not use independent supervision for the LR or MR stages, nor does it adopt sequential training, alternating optimization, or stagewise parameter freezing. The training loss is imposed only on the final HR reconstruction, and the detailed loss function is introduced later.

C. Observation-Consistency-Guided MR Unfolding

The MR stage is the main refinement stage of PST-DUNet. Let $\mathbf{Z}^{(k-1)} \in \mathbb{R}^{2h\times 2w\times B}$ denote the MR image-domain estimate from the $(k-1)$th unfolding step, and let $\mathbf{Y} \in \mathbb{R}^{h\times w\times B}$ denote the observed LR HSI.

The MR unfolding step is motivated by the following observation-consistency regularized reconstruction form:

$$\min_{\mathbf{Z}} \frac{1}{2}\|D_m(\mathbf{Z}) - \mathbf{Y}\|_F^2 + \lambda_m \Phi_\theta(\mathbf{Z}) \tag{2}$$

where $D_m(\cdot)$ denotes the MR-to-LR observation projection and represents an implicit learned prior parameterized by neural network modules. This formulation is used only as an optimization-inspired motivation. PST-DUNet does not explicitly solve a convex variational problem, nor does it define a closed-form proximal operator for Φ_θ.

For the observation-consistency correction, the DC module constructs a learnable MR-to-LR projection and a learnable residual back-projection

$$D_m(\mathbf{Z}) = \mathrm{DWConv}_{s_m}(\mathbf{Z}; \Theta_D) \tag{3}$$

$$B_m(\mathbf{R}) = \mathrm{DWTransConv}_{s_m}(\mathbf{R}; \Theta_B) \tag{4}$$

where $s_m = 2$ is the MR-to-LR scale factor, $\mathrm{DWConv}_{s_m}(\cdot)$ denotes depthwise strided convolution with stride s_m, and $\mathrm{DWTransConv}_{s_m}(\cdot)$ denotes depthwise transposed convolution with stride s_m. Θ_D and Θ_B are learnable parameters. $\mathbf{R}$ is the correction result in LR space. The depthwise design makes the DC operation spatial and bandwise, without introducing additional spectral mixing inside the DC module.

Based on these two operators, the DC correction in the kth unfolding step is defined as

$$\mathbf{Z}_{DC}^{(k)} = \mathbf{Z}^{(k-1)} - \eta_k B_m\left(D_m\left(\mathbf{Z}^{(k-1)}\right) - \mathbf{Y}\right) \tag{5}$$

where $\mathbf{Z}_{DC}^{(k)}$ denotes the DC-corrected MR estimate and η_k is a learnable step size. The residual term $D_m(\mathbf{Z}^{(k-1)}) - \mathbf{Y}$ measures the inconsistency between the projected MR estimate and the observed LR HSI.

It should be noted that this update is a parameterized observation-consistency correction inspired by the data-fidelity gradient step. It is not an exact gradient descent step with a fixed degradation operator. In a strict gradient step for $1/2\|D_m(\mathbf{Z}) - \mathbf{Y}\|_F^2$, the correction direction would involve the adjoint of the forward observation operator. In PST-DUNet, $B_m(\cdot)$ is implemented as a learnable residual back-projection operator, allowing the network to learn how the LR residual should be distributed back to the MR space under the reconstruction objective.

After DC correction, PST-DUNet performs learned prior refinement. In particular, DSTI infers reconstruction-state-conditioned interband dependencies from $Z_{DC}^{(k)}$, and STR transforms the topology-aggregated spectral representation into feature modulation for spatial refinement. The updated MR estimate is written as

$$f^{(k)} = \Psi_\theta^{(k)}\left(f^{(k-1)}, \mathbf{Z}_{DC}^{(k)}, \mathbf{T}^{(k)}\right) \tag{6}$$

$$\mathbf{Z}^{(k)} = \mathbf{Z}_{DC}^{(k)} + \psi_{\mathrm{img}}^{(k)}\left(f^{(k)}\right) \tag{7}$$

where $\mathbf{T}^{(k)}$ denotes the topology-aggregated spectral representation generated from the current MR estimate, $\Psi_\theta^{(k)}(\cdot)$ denotes the learned feature refinement module, and $\psi_{\mathrm{img}}^{(k)}(\cdot)$ maps the refined feature to an MR residual image. The detailed definitions of DSTI and STR are provided in Sections III-D and III-E, respectively.

By combining the observation-consistency correction and the learned prior refinement, the kth MR unfolding step can be compactly expressed as

$$\mathbf{Z}^{(k)} = \mathcal{U}_\theta^{(k)}\left(\mathbf{Z}^{(k-1)}, \mathbf{Y}\right), \quad k = 1, \ldots, K \tag{8}$$

where $\mathcal{U}_\theta^{(k)}(\cdot)$ denotes the learned update function implemented by the DC module, DSTI, STR, and spatial feature refinement with convolutional residual blocks. After K MR unfolding steps, the refined MR estimate $\mathbf{Z}^{(K)}$ is fed into the subsequent HR reconstruction module. We set $K = 3$ in PST-DUNet.

D. Instance-Adaptive DSTI

The DSTI module is designed to infer reconstruction-state-conditioned interband dependencies from the current image-domain estimate. Let $\mathbf{I}$ denote the image-domain input of DSTI. In PST-DUNet, $\mathbf{I}$ has different meanings in different reconstruction stages

$$\mathbf{I} = \begin{cases} \mathbf{Y}, & \text{for the LR-space DSTI input} \\ \mathbf{Z}_{DC}^{(k)}, & \text{for the } k\text{-th MR unfolding step.} \end{cases} \tag{9}$$

In this way, DSTI constructs the interband dependency representation according to the current input or reconstruction state, rather than using a fixed spectral graph shared by all inputs and all unfolding steps.

We construct a dynamic spectral graph whose nodes correspond to spectral bands. Thus, there are B band nodes for an HSI with B spectral bands. For each band node, DSTI first extracts a band-centered descriptor from $\mathbf{I}$. In particular, we use 3-D convolutions to encode local spatial–spectral context around each band, yielding band-indexed feature responses

$$\mathbf{F} = \mathrm{Conv3D}(\mathbf{I}) \in \mathbb{R}^{B\times C_f\times \hat{h}\times \hat{w}}, \quad \mathbf{F}_b \in \mathbb{R}^{C_f\times \hat{h}\times \hat{w}} \tag{10}$$

where F_b denotes the feature response associated with the bth band node and C_f represents the number of feature channels.

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To summarize the spatial distribution of each band response, we perform attention-weighted spatial aggregation

$$\alpha_b = \mathrm{softmax}\left(\mathrm{Conv2D}\left(\mathbf{F}_b\right)\right), \quad \mathbf{z}_b^{att} = \sum_p \alpha_b(p)\,\mathbf{F}_b(p) \tag{11}$$

where p indexes spatial positions. To preserve complementary response statistics, global maximum pooling and standard deviation statistics are also used. The band descriptor is then obtained by

$$\mathbf{z}_b = \left[\mathbf{z}_b^{att}, \mathbf{z}_b^{max}, \mathbf{z}_b^{std}\right]. \tag{12}$$

Then, the descriptor is mapped into the node feature space, and a learnable band-index embedding is added

$$\mathbf{h}_b = \psi\left(\mathbf{z}_b\right) + \mathbf{e}_b \in \mathbb{R}^d, \quad \mathbf{H}(\mathbf{I}) = [\mathbf{h}_1, \ldots, \mathbf{h}_B]^\top \in \mathbb{R}^{B\times d} \tag{13}$$

where d denotes the node descriptor dimension and $\mathbf{e}_b$ denotes the learnable embedding for the bth band index.

Based on the band-node feature matrix $\mathbf{H}$, DSTI constructs a dynamic spectral adjacency matrix through learnable query-key projections

$$\mathbf{A}(\mathbf{I}) = \mathrm{softmax}\left(\frac{\left(\mathbf{H}(\mathbf{I})\,W_Q\right)\left(\mathbf{H}(\mathbf{I})\,W_K\right)^\top}{\tau\sqrt{d}}\right) \tag{14}$$

where W_Q and W_K are learnable projection matrices and τ is a learnable temperature coefficient. The softmax operation is performed rowwise.

More explicitly, the edge weight from the jth band node to the ith band node is defined as

$$a_{ij}(\mathbf{I}) = \frac{\exp\left(\frac{\left(\mathbf{h}_i(\mathbf{I})W_Q\right)\left(\mathbf{h}_j(\mathbf{I})W_K\right)^\top}{\tau\sqrt{d}}\right)}{\sum_{l=1}^{B}\exp\left(\frac{\left(\mathbf{h}_i(\mathbf{I})W_Q\right)\left(\mathbf{h}_l(\mathbf{I})W_K\right)^\top}{\tau\sqrt{d}}\right)}. \tag{15}$$

The element $a_{ij}(\mathbf{I})$ indicates the contribution weight of the jth band node when updating the ith band node under the current reconstruction state.

Finally, based on the inferred adjacency matrix, DSTI performs attention-style spectral node aggregation

$$\mathbf{T}(\mathbf{I}) = \mathbf{A}(\mathbf{I})\left(\mathbf{H}(\mathbf{I})\,W_V\right) \tag{16}$$

where W_V is the learnable value projection matrix, and $\mathbf{T}(\mathbf{I}) \in \mathbb{R}^{B\times d}$ denotes the topology-aggregated spectral node representation. This representation is then passed to STR to generate spectral-dependency-conditioned modulation for the spatial feature update.

E. Spectral Topology Rectifier

The DSTI module produces a topology-aggregated spectral node representation $\mathbf{T}(\mathbf{I}) \in \mathbb{R}^{B\times d}$ from the current image-domain state $\mathbf{I}$. To make this interband dependency representation participate in spatial detail recovery, we introduce the STR. STR transforms $\mathbf{T}(\mathbf{I})$ into channelwise modulation coefficients and uses them to condition the spatial feature update.

Let $\mathbf{F}_{\mathrm{spa}} \in \mathbb{R}^{C_f\times H_s\times W_s}$ denote the spatial feature to be updated at the current reconstruction stage, which is calculated by

$$\mathbf{F}_{\mathrm{spa}} = f^{(k-1)} + \psi_{feat}\left(\mathbf{Z}_{DC}^{(k)} - \mathbf{Z}^{(k-1)}\right) \tag{17}$$

where $f^{(k-1)}$ denotes the refined MR feature representation from the previous unfolding stage and $\psi_{feat}(\cdot)$ represents the image-to-feature mapping. STR first applies a spatial enhancement module, which has multiple convolutional residual blocks, to obtain

$$\tilde{\mathbf{F}}_{\mathrm{spa}} = \mathcal{S}\left(\mathbf{F}_{\mathrm{spa}}\right) \tag{18}$$

where $\mathcal{S}(\cdot)$ is implemented by residual convolutional blocks. The local textures, edges, and spatial structures are mainly modeled in this spatial enhancement branch.

Then, STR performs a learnable graph readout on the topology-aggregated spectral node representation. In particular, the spectral condition vector is obtained as

$$\hat{\mathbf{t}} = \sum_{b=1}^{B}\omega_b\mathbf{T}_b(\mathbf{I}) \tag{19}$$

where $T_b(I) \in \mathbb{R}^d$ denotes the bth topology-aggregated band-node representation, and ω_b is a learnable readout weight for the bth band node. The spectral condition vector $\hat{t}$ summarizes the interband dependency information inferred from the current reconstruction state. Based on $\hat{t}$, STR generates a channelwise gate by

$$g = \sigma\left(\mathrm{MLP}\left(\hat{\mathbf{t}}\right)\right) \tag{20}$$

where $g \in \mathbb{R}^{C_f}$, $\sigma(\cdot)$ denotes the sigmoid function, and the MLP maps the spectral condition vector to the channel dimension of the spatial feature. The modulated output is then computed as

$$\mathbf{F}_{\mathrm{out}} = \tilde{\mathbf{F}}_{\mathrm{spa}} \odot g \tag{21}$$

where $\odot$ denotes channelwise multiplication. Therefore, the gate generated by STR is channelwise and spatially shared, rather than a pixelwise spatial attention map.

Different from standard channel attention, whose gate is usually generated from pooled statistics of the feature to be modulated, STR conditions the spatial feature update on the topology-aggregated spectral representation produced by DSTI. In this way, the recovery of spatial details is coordinated with the current interband dependency representation. Meanwhile, spatial variations such as boundaries and local textures are still modeled by the convolutional spatial enhancement branch $\mathcal{S}(\cdot)$.

F. Loss Function

PST-DUNet is trained in an end-to-end manner using the final HR reconstruction as supervision. Given the reconstructed HR-HSI $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$ and the reference HR-HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, the training objective consists of a reconstruction loss, a spectral angular loss, and a spatio-spectral gradient loss

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda_{sam}\mathcal{L}_{sam} + \lambda_{grad}\mathcal{L}_{grad}. \tag{22}$$

The reconstruction loss is defined as the pixelwise ℓ_1 loss

$$\mathcal{L}_{rec} = \frac{1}{BHW}\left\|\hat{\mathbf{X}} - \mathbf{X}\right\|_1. \tag{23}$$

Compared with the ℓ_2 loss, the ℓ_1 loss is less sensitive to large local errors and helps alleviate oversmoothed reconstruction results.

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To encourage spectral angular consistency, we use the SAM loss

$$\mathcal{L}_{sam} = \frac{1}{HW}\sum_{p=1}^{HW}\arccos\left(\frac{\langle\hat{\mathbf{x}}_p, \mathbf{x}_p\rangle}{\|\hat{\mathbf{x}}_p\|_2 \|\mathbf{x}_p\|_2 + \epsilon}\right) \tag{24}$$

where $\hat{\mathbf{x}}_p$ and $\mathbf{x}_p$ denote the reconstructed and reference spectral vectors at spatial position p, respectively, and ϵ is a small constant for numerical stability.

In addition, we impose gradient consistency along both spatial and spectral directions

$$\mathcal{L}_{grad} = \left\|\nabla_h\hat{\mathbf{X}} - \nabla_h\mathbf{X}\right\|_1 + \left\|\nabla_w\hat{\mathbf{X}} - \nabla_w\mathbf{X}\right\|_1 + \left\|\nabla_b\hat{\mathbf{X}} - \nabla_b\mathbf{X}\right\|_1 \tag{25}$$

where $\nabla_h(\cdot)$, $\nabla_w(\cdot)$, and $\nabla_b(\cdot)$ denote the first-order gradients along the height, width, and spectral dimensions, respectively.

In this article, we set $\lambda_{sam} = 0.4$ and $\lambda_{grad} = 0.1$, and keep these weights fixed across all datasets and scale factors. No additional LR-level or MR-level supervision is imposed. The LR observation consistency is introduced through the DC module in the forward reconstruction process, rather than through an extra loss term.

IV. Experiment

In this section, we evaluate the proposed PST-DUNet on several publicly available hyperspectral datasets. We first introduce the experimental settings, including the datasets, data preparation protocol, training details, and evaluation metrics. Then, we compare PST-DUNet with representative HSI-SR methods under the same experimental settings. Finally, ablation studies and sensitivity analyses are conducted to evaluate the contributions of the key components to spatial detail recovery and spectral consistency preservation.

A. Experimental Settings

1) Datasets: We evaluate the proposed PST-DUNet on three representative hyperspectral datasets: Pavia Center, Chikusei, and CAVE. Pavia Center and Chikusei are remote-sensing HSI datasets acquired from real-world airborne imaging platforms, which contain practical factors such as mixed pixels, spatial heterogeneity, and sensor-related artifacts. CAVE is an indoor HSI dataset collected under controlled illumination and imaging conditions. It provides high-quality spectral images and serves as a complementary benchmark for evaluating reconstruction behavior under cleaner imaging conditions.

Pavia Center: This dataset was acquired by the ROSIS sensor over Pavia, Italy. The original image has a spatial resolution of 1096×1096 with 102 spectral bands. Since the effective area only covers a 1096×715 region and contains a distinct discontinuous boundary, we first removed the uninformative areas. The remaining effective region was then divided along the boundary into two parts: the right part for training and the left part for testing.

Chikusei: The Chikusei dataset was captured by the Hyperspec-VNIR-C sensor in Chikusei, Japan, with a spatial dimension of 2517×2335 and 128 spectral bands. Considering the information loss at the edges, we cropped an effective area of 2304×2048 from the center. This region was further partitioned, with the upper portion used for testing and the lower portion for training.

CAVE: This dataset consists of 32 HSIs of indoor scenes. Each image has a spatial resolution of 512×512 across 31 spectral bands. Following the common experimental setup, six images were selected for testing, while the remaining 26 images were used for training.

Sample Preparation: To construct the training and testing sets, we adopt a nonoverlapping cropping strategy. During training, HR patches with the size of 128×128 are uniformly cropped from the training regions. For evaluation, the testing patch sizes are adjusted according to the dataset characteristics: 128×128 for Pavia Center, 256×256 for Chikusei, and the full 512×512 resolution for CAVE. For all datasets and scale factors, the LR observations are generated from the corresponding HR-HSIs by bicubic downsampling. The downsampling operation is applied band by band along the spatial dimensions and does not change the spectral dimension. No additional noise is added during paired data construction. Therefore, the experiments follow a known nonblind degradation setting. We perform experiments under $\times 4$ and $\times 8$ scale factors.

2) Training Setting: We trained our network using the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. The initial learning rate was set to 10^{-4} and decayed by a factor of 10 every 70 epochs during training. All experiments were implemented in PyTorch and trained on an NVIDIA RTX 4090 GPU.

3) Evaluation Metrics: We evaluate the reconstruction quality using six widely adopted metrics in HSI-SR, including mean peak signal-to-noise ratio (MPSNR), mean structural similarity index (MSSIM), SAM, root mean square error (RMSE), erreur relative globale adimensionnelle de synthése (ERGAS), and correlation coefficient (CC). Let $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ denote the reconstructed and reference HR-HSIs, respectively. Let $\hat{\mathbf{X}}_b$ and $\mathbf{X}_b$ denote the bth reconstructed and reference band images.

First, MPSNR is computed by averaging the PSNR values over all spectral bands

$$\text{MPSNR} = \frac{1}{B}\sum_{b=1}^{B} 10\log_{10}\left(\frac{L^2}{\text{mse}_b}\right) \tag{26}$$

where L denotes the maximum possible pixel value and

$$\text{mse}_b = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\left(\hat{\mathbf{X}}_{b,u,v} - \mathbf{X}_{b,u,v}\right)^2. \tag{27}$$

A higher MPSNR indicates better pixelwise reconstruction fidelity.

Second, RMSE measures the average magnitude error over all pixels and bands

$$\text{RMSE} = \sqrt{\frac{1}{BHW}\sum_{b=1}^{B}\sum_{u=1}^{H}\sum_{v=1}^{W}\left(\hat{\mathbf{X}}_{b,u,v} - \mathbf{X}_{b,u,v}\right)^2}. \tag{28}$$

A lower RMSE indicates smaller spectral–spatial magnitude error.

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Third, SAM measures the spectral angular deviation between reconstructed and reference spectral vectors

$$\mathrm{SAM} = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\arccos\left(\frac{\langle\hat{\mathbf{x}}_{u,v},\mathbf{x}_{u,v}\rangle}{\|\hat{\mathbf{x}}_{u,v}\|_2\|\mathbf{x}_{u,v}\|_2+\epsilon}\right) \tag{29}$$

where $\hat{\mathbf{x}}_{u,v}$ and $\mathbf{x}_{u,v}$ denote the reconstructed and reference spectral vectors at spatial position (u,v), respectively, and ϵ is a small constant for numerical stability. A lower SAM indicates better spectral shape consistency.

Fourth, MSSIM is computed by averaging SSIM over all spectral bands

$$\mathrm{MSSIM} = \frac{1}{B}\sum_{b=1}^{B}\mathrm{SSIM}\left(\hat{\mathbf{X}}_b,\mathbf{X}_b\right). \tag{30}$$

For each band, SSIM is defined as

$$\mathrm{SSIM}\left(\hat{\mathbf{X}}_b,\mathbf{X}_b\right) = \frac{\left(2\mu_{\hat{\mathbf{X}}_b}\mu_{\mathbf{X}_b}+C_1\right)\left(2\sigma_{\hat{\mathbf{X}}_b\mathbf{X}_b}+C_2\right)}{\left(\mu_{\hat{\mathbf{X}}_b}^2+\mu_{\mathbf{X}_b}^2+C_1\right)\left(\sigma_{\hat{\mathbf{X}}_b}^2+\sigma_{\mathbf{X}_b}^2+C_2\right)} \tag{31}$$

where $\mu_{\hat{\mathbf{X}}_b}$ and $\mu_{\mathbf{X}_b}$ denote the mean values, $\sigma^2_{\hat{\mathbf{X}}_b}$ and $\sigma^2_{\mathbf{X}_b}$ denote the variances, $\sigma_{\hat{\mathbf{X}}_b\mathbf{X}_b}$ denotes the covariance, and C_1 and C_2 are stabilizing constants. A higher MSSIM indicates better spatial structural similarity.

Fifth, ERGAS measures the relative global reconstruction error

$$\mathrm{ERGAS} = \frac{100}{r}\sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\frac{\mathrm{RMSE}_b}{\mu_{\mathbf{X}_b}}\right)^2} \tag{32}$$

where r denotes the scale factor, $\mu_{\mathbf{X}_b}$ is the mean value of the bth reference band, and

$$\mathrm{RMSE}_b = \sqrt{\frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\left(\hat{\mathbf{X}}_{b,u,v}-\mathbf{X}_{b,u,v}\right)^2}. \tag{33}$$

A lower ERGAS indicates smaller relative global error.

Finally, CC measures the correlation between the reconstructed and reference HSIs. We compute the bandwise CC and average it over all bands

$$\begin{aligned}
\mathrm{CC} &= \frac{1}{B}\sum_{b=1}^{B}\frac{N_b}{\sqrt{D_{\hat{\mathbf{X}},b}}\sqrt{D_{\mathbf{X},b}}}\\
N_b &= \sum_{u=1}^{H}\sum_{v=1}^{W}\left(\hat{\mathbf{X}}_{b,u,v}-\mu_{\hat{\mathbf{X}}_b}\right)\left(\mathbf{X}_{b,u,v}-\mu_{\mathbf{X}_b}\right)\\
D_{\hat{\mathbf{X}},b} &= \sum_{u=1}^{H}\sum_{v=1}^{W}\left(\hat{\mathbf{X}}_{b,u,v}-\mu_{\hat{\mathbf{X}}_b}\right)^2\\
D_{\mathbf{X},b} &= \sum_{u=1}^{H}\sum_{v=1}^{W}\left(\mathbf{X}_{b,u,v}-\mu_{\mathbf{X}_b}\right)^2.
\end{aligned} \tag{34}$$

A higher CC indicates stronger correlation with the reference HSI.

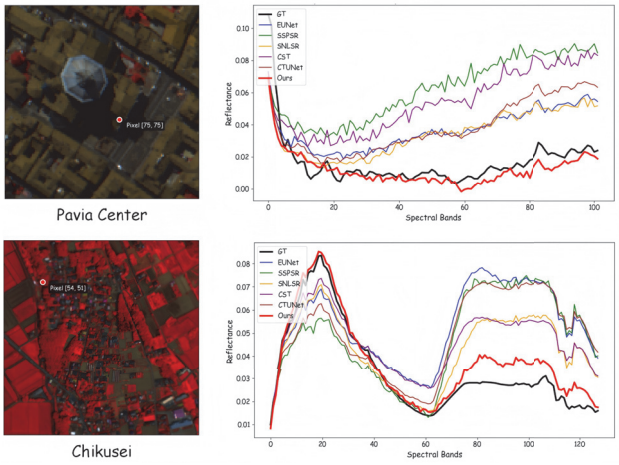


Fig. 3. Comparison of spectral curves of reconstructed results. Our PST-DUNet (red line) performs the best reconstruction to the ground truth (black line).

B. Comparisons With the State-of-the-Art Methods

We compared the proposed PST-DUNet with a set of representative super-resolution methods on all three datasets. In particular, we included two general image SR baselines, RCAN [44] and RDN [45], as well as several HSI-oriented approaches, including GDRNN [34], EUNet [41], CTUNet [43], SSPSR [35], SNLSR [37], and CST [39]. For a fair comparison, all competing methods were implemented using their publicly available code, which was trained and evaluated under the same settings as PST-DUNet.

1) Experiments on the Real-Scenario HSIs: We conduct $\times 4$ and $\times 8$ super-resolution experiments on Pavia Center and Chikusei. Quantitative results are summarized in Tables I and II, while spectral profiles and visual comparisons of error maps are presented in Figs. 3–5. Overall, PST-DUNet achieves superior performance under the adopted nonblind bicubic degradation setting. The improvements are observed not only in spatial fidelity metrics such as MPSNR and MSSIM but also in spectral consistency metrics such as SAM and RMSE.

To elucidate the performance discrepancies, we analyze the core design of each method. Natural-image SR methods like RCAN and RDN primarily establish direct LR–HR mappings for spatial texture reconstruction, assuming that channel dimensions can be freely reorganized in the feature domain. However, in HSIs, channels correspond to physical bands, and cross-band consistency is an indispensable constraint. In real remote-sensing scenarios where spatial cues are inherently deficient, these methods struggle with inconsistent interband responses, leading to a significant lag in spectral fidelity metrics compared to HSI-specific approaches.

Among HSI-specific methods, GDRNN enhances representation through deep recursive residuals and grouped convolutions, but its limited exploitation of spectral structural priors causes it to reach a performance bottleneck under severe degradation. EUNet embeds the reconstruction within a model-driven iterative framework with explicit degradation consistency. However, its prior module relies on spatial–spectral decoupled convolutions and local spectral

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TABLE I
EXPERIMENTAL RESULTS ON PAVIA CENTER DATASET

	Metrics	RCAN [46]	RDN [47]	GDRRN [36]	EUNet [43]	CTUNet [45]	SSPSR [37]	SNLSR [39]	CST [41]	Ours
Pavia×4	MPSNR↑	28.3220	26.9618	27.7656	28.8874	29.0195	28.4184	28.9305	28.8161	**29.3289**
	MSSIM↑	0.7676	0.7223	0.7319	0.7925	0.7954	0.7711	0.7953	0.7884	**0.8120**
	SAM↓	7.0514	10.4168	7.5645	5.7410	5.8192	6.4544	5.9456	6.0168	**5.4822**
	RMSE↓	0.0398	0.0462	0.0426	0.0373	0.0367	0.0395	0.0371	0.0376	**0.0354**
	ERGAS↓	6.5545	7.7573	6.9667	6.1309	6.0459	6.4653	6.1099	6.1885	**5.8374**
	CC↑	0.8629	0.8162	0.8443	0.8811	0.8851	0.8669	0.8818	0.8787	**0.8925**
Pavia×8	MPSNR↑	25.0778	24.4615	24.9774	25.3620	25.3253	25.3172	25.1717	25.2986	**25.4703**
	MSSIM↑	0.5032	0.4839	0.4892	0.5260	0.5143	0.5192	0.5116	0.5188	**0.5357**
	SAM↓	8.4361	11.5705	8.7208	7.3689	7.5330	7.7819	7.6047	7.7323	**7.2981**
	RMSE↓	0.0593	0.0632	0.0600	0.0574	0.0578	0.0578	0.0586	0.0579	**0.0567**
	ERGAS↓	9.4101	10.1509	9.5210	9.1101	9.1424	9.1492	9.3173	9.1714	**8.9947**
	CC↑	0.6910	0.6511	0.6816	0.7128	0.7101	0.7093	0.6982	0.7083	**0.7211**

TABLE II
EXPERIMENTAL RESULTS ON CHIKUSEI DATASET

	Metrics	RCAN [46]	RDN [47]	GDRRN [36]	EUNet [43]	CTUNet [45]	SSPSR [37]	SNLSR [39]	CST [41]	Ours
Chikusei×4	MPSNR↑	39.9177	41.1650	41.7293	42.3916	42.6070	42.2814	42.5307	42.7935	**42.8840**
	MSSIM↑	0.9322	0.9469	0.9493	0.9549	0.9572	0.9539	0.9564	0.9587	**0.9595**
	SAM↓	3.2305	2.7804	2.0918	1.8053	1.7734	1.9318	1.7824	1.7396	**1.7005**
	RMSE↓	0.0122	0.0107	0.0104	0.0098	0.0095	0.0099	0.0096	0.0094	**0.0093**
	ERGAS↓	5.9710	5.1618	6.6746	4.2765	4.2012	4.3524	4.2391	4.1054	**4.0591**
	CC↑	0.9198	0.9378	0.9473	0.9544	0.9561	0.9531	0.9555	0.9579	**0.9586**
Chikusei×8	MPSNR↑	36.9916	37.0640	37.5941	37.9556	38.0674	38.1735	38.0322	38.1890	**38.3206**
	MSSIM↑	0.8648	0.8728	0.8794	0.8856	0.8850	0.8889	0.8858	0.8885	**0.8919**
	SAM↓	4.0891	4.3360	3.2989	3.9977	3.0569	3.0538	3.0218	3.0162	**2.8969**
	RMSE↓	0.0177	0.0174	0.0169	0.0163	0.0162	0.0160	0.0163	0.0160	**0.0158**
	ERGAS↓	8.0357	8.0826	7.3926	6.9571	6.8806	6.8263	6.9412	6.8104	**6.6831**
	CC↑	0.8449	0.8496	0.8648	0.8767	0.8788	0.8818	0.8774	0.8820	**0.8848**

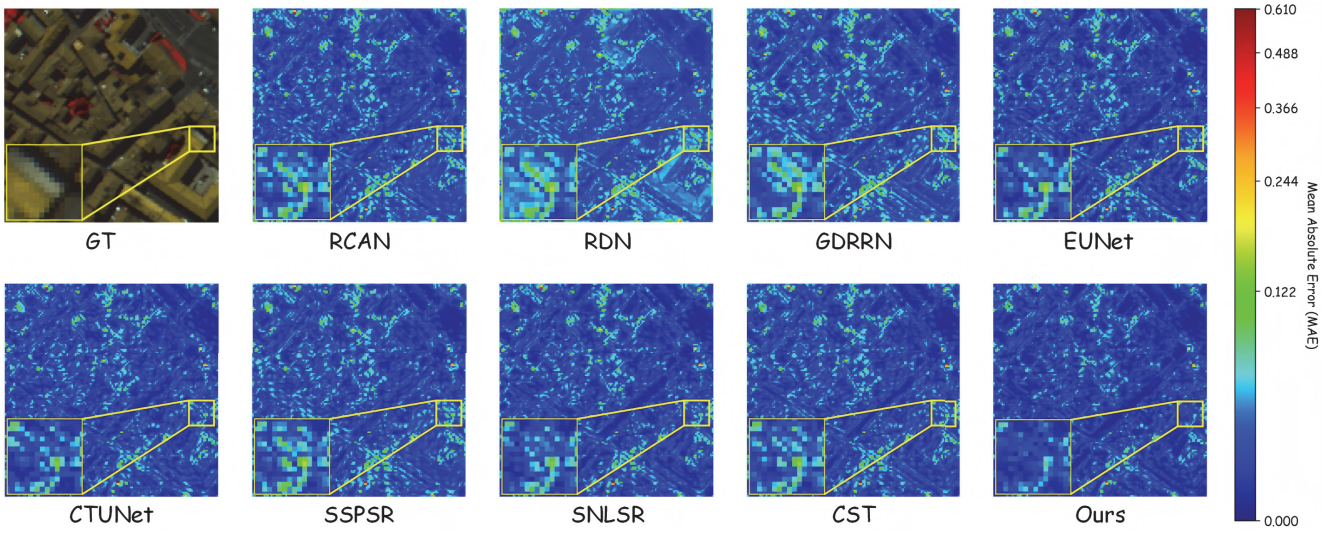


Fig. 4. Mean absolute error visualization results on ×4 Pavia Center.

injection. Since interband dependencies in real-world data often exhibit nonlocal and long-range characteristics, EUNet's local spectral modeling remains suboptimal for high-scale SR in complex regions.

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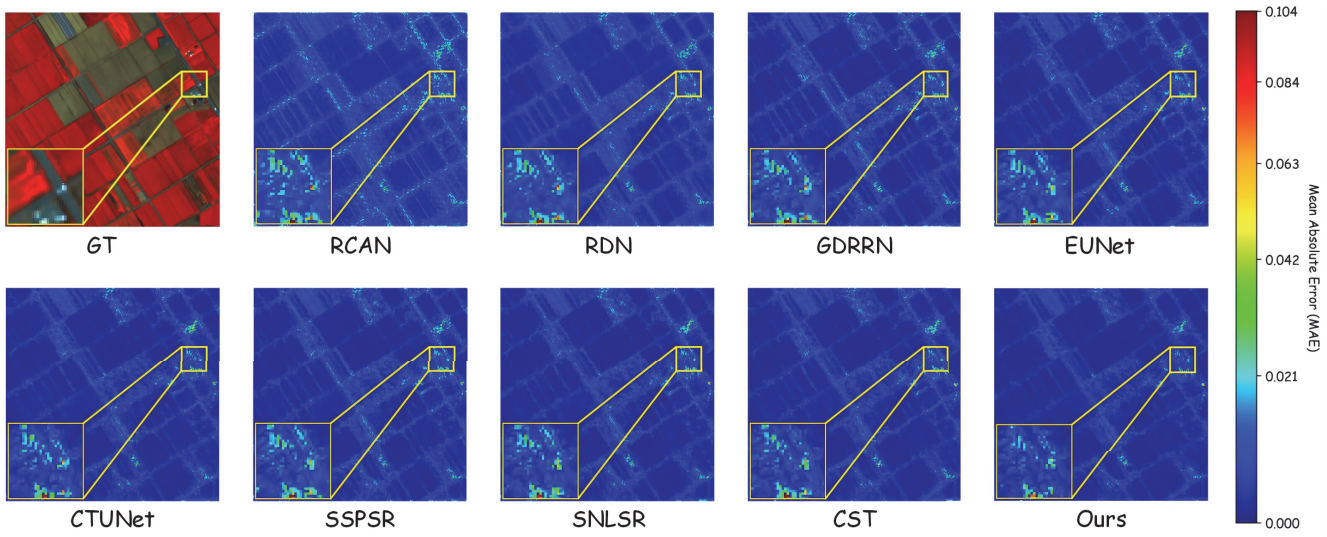


Fig. 5. Mean absolute error visualization results on $\times 4$ Chikusei.

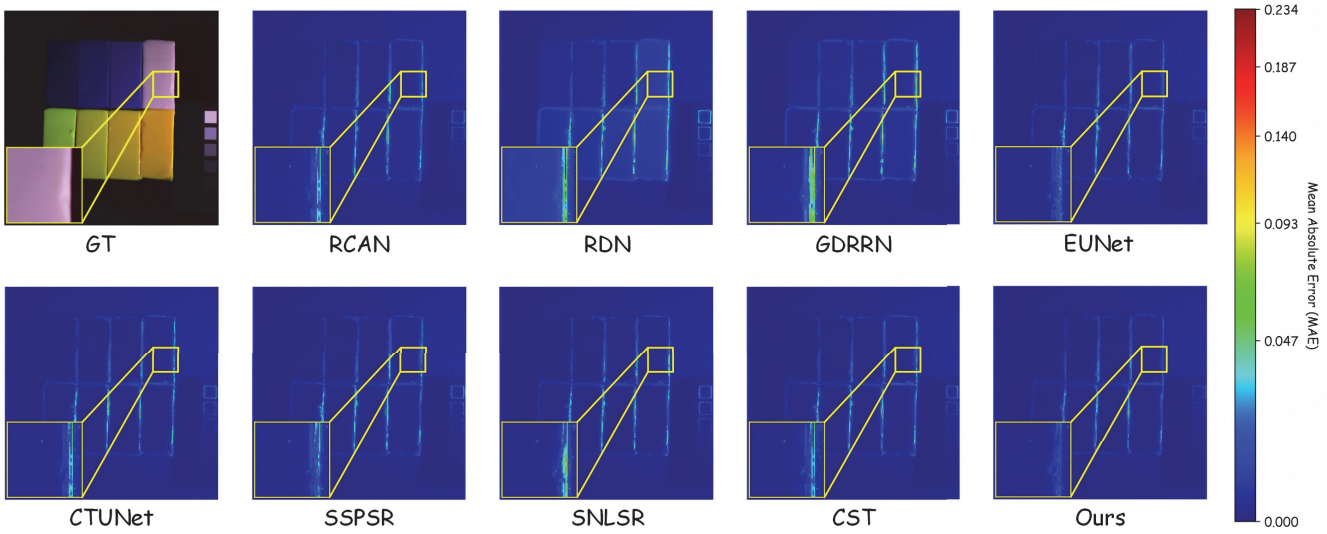


Fig. 6. Mean absolute error visualization results on $\times 4$ CAVE.

CTUNet also leverages an unfolding paradigm with learnable spatial–spectral priors. While its integration of Transformer mechanisms and edge-guided attention strengthens nonlocal information exploitation, the spectral consistency constraint remains implicitly manifested through feature-domain interactions. Consequently, when substantial spatial detail recovery is required, shifts in cross-band responses may still occur.

Furthermore, SSPSR utilizes hand-crafted band grouping for local reconstruction followed by global fusion. Although this maintains spatial recovery at high scaling factors, the grouping strategy weakens long-range interband correlations and assumes similar statistical properties across bands, which is often violated in heterogeneous remote-sensing scenes. SNLSR operates in the abundance domain to model long-range correlations in a low-dimensional space. However, its correlation modeling is performed on learned representations rather than the original physical band space, leading to a structural discrepancy. CST utilizes global attention to enhance nonlocal modeling, but its performance is highly sensitive to data scale and distribution, which limits its robustness in complex scenarios with limited samples.

2) Experiments on Synthesized HSIs: To further validate the reconstruction performance of PST-DUNet under ideal imaging conditions, we conducted comparative experiments on the CAVE dataset. Quantitative evaluation metrics for scaling factors of $\times 4$ and $\times 8$ are summarized in Table III. Furthermore, to visually contrast the discrepancies in detail recovery among different methods, the error maps of the reconstruction results are presented in Fig. 6.

Compared with the SR results on Pavia Center and Chikusei, all competitive methods exhibit a significant improvement in numerical metrics on the CAVE dataset. This is primarily attributed to the unique imaging characteristics of the CAVE data: as synthetic data acquired in a controlled laboratory environment, it possesses a higher signal-to-noise ratio (SNR) and

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TABLE III
EXPERIMENTAL RESULTS ON CAVE DATASET

	Metrics	RCAN [46]	RDN [47]	GDRRN [36]	EUNet [43]	CTUNet [45]	SSPSR [37]	SNLSR [39]	CST [41]	Ours
CAVE×4	MPSNR↑	40.3800	38.5891	39.2065	40.7176	40.8562	40.8674	40.7805	41.0600	**41.2670**
	MSSIM↑	0.9758	0.9711	0.9716	0.9773	0.9779	0.9779	0.9781	0.9784	**0.9794**
	SAM↓	3.6867	4.4734	3.9626	3.3175	3.3141	3.4849	3.3163	3.2837	**3.2174**
	RMSE↓	0.0106	0.0129	0.0118	0.0101	0.0099	0.0099	0.0100	0.0097	**0.0095**
	ERGAS↓	5.1788	6.1557	5.7737	4.9767	4.8449	5.0031	4.9464	4.8421	**4.7527**
	CC↑	0.9914	0.9881	0.9895	0.9918	0.9924	0.9919	0.9921	0.9924	**0.9926**
CAVE×8	MPSNR↑	36.1646	34.7895	34.9378	36.4564	36.3191	36.5084	36.5382	36.5480	**36.6283**
	MSSIM↑	0.9492	0.9403	0.9426	0.9535	0.9516	0.9530	0.9540	0.9531	**0.9548**
	SAM↓	4.8251	5.9699	4.9450	4.0551	4.1253	4.3152	4.0977	4.0284	**3.8916**
	RMSE↓	0.0169	0.0196	0.0193	0.0163	0.0165	0.0163	0.0162	0.0162	**0.0161**
	ERGAS↓	7.8205	8.9358	8.7841	7.5501	7.6758	7.6027	7.5123	7.5116	**7.5068**
	CC↑	0.9802	0.9743	0.9755	0.9809	0.9803	0.9807	0.9810	0.9812	**0.9812**

TABLE IV
ABLATION EXPERIMENTS ON 4× PAVIA CENTER DATASET

Experiment number	Network Setting	MPSNR↑	MSSIM↑	SAM↓	RMSE↓	ERGAS↓	CC↑
I	PST-DUNet	**29.3289**	**0.8120**	**5.4822**	**0.0354**	**5.8374**	**0.8925**
II	Unfold at HR-space	29.0200	0.7999	5.5147	0.0367	6.0443	0.8845
III	LR-HR mapping w/o STR	28.9837	0.7960	5.8007	0.0369	6.0682	0.8833
IV	w/o STR	29.2562	0.8084	5.6444	0.0357	5.8829	0.8908
V	Concat fusion	28.9891	0.7976	5.8676	0.0368	6.0686	0.8844
VI	w/o Temp	29.2567	0.8091	5.5648	0.0357	5.8938	0.8907
VII	w/o DSTI (self-loop)	29.3064	0.8108	5.5327	0.0356	5.8460	0.8922
VIII	Fixed full graph	29.2750	0.8100	5.5966	0.0357	5.8710	0.8913

largely consists of individual indoor objects with exceptionally sharp edge contours. In addition, the degradation simulation process for this dataset is fully controllable, circumventing the unstable factors inherent in real-world remote-sensing imaging.

It is evident that PST-DUNet maintains a distinct leading advantage across all key evaluation metrics. Analysis combined with the error maps indicates that for indoor objects with high-contrast boundaries, while RCAN and RDN manage to recover sharp contours, they still exhibit noticeable artifacts in the error maps. Meanwhile, although HSI-oriented SR methods achieve competitive numerical results, they still struggle to model more authentic spectral relationships.

These results indicate that PST-DUNet can effectively coordinate spatial detail recovery and spectral consistency preservation on the evaluated datasets. The advantage mainly comes from the progressive MR-space refinement and the reconstruction-state-conditioned interband dependency modulation introduced by DSTI and STR.

C. Ablation Experiments

To verify the effectiveness of the key designs in PST-DUNet and their contributions to the final reconstruction performance, we conducted a set of ablation experiments on the Pavia Center dataset for the ×4 SR. The quantitative results for various model variants are summarized in Table IV. To comprehensively evaluate the impact of each component, we designed six different experimental configurations: Experiment I PST-DUNet denotes the proposed full model; Experiment II (Unfold at HR-space) degrades the model by performing deep unfolding directly in the HR space, thereby removing the intermediate processing at the MR scale; Experiment III (LR–HR mapping w/o STR) further removes the spectral branch and STR, retaining only the spatial branch to learn a direct LR–HR mapping; Experiment IV (w/o STR) removes the spectral branch from PST-DUNet, relying solely on the spatial branch for iterative updates; Experiment V (Concat Fusion) retains the spectral branch but replaces the conditional gating mechanism of the STR with channel concatenation; and Experiment VI (w/o Temp) removes the learnable temperature coefficient τ from the DSTI module; Experiment VII inpalces the dynamic interband dependency inference with self-loop on every band-node; and Experiment VIII inpalces the dynamic interband dependency inference with a fixed full graph.

Based on Table IV, we analyze the contribution of PST-DUNet from four aspects.

1) Contribution of MR-Space Progressive Refinement: Comparing Experiment I with Experiments II and III shows the effect of the progressive MR-space reconstruction path.

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TABLE V
SENSITIVITY ANALYSIS OF LOSS WEIGHTS ON ×4 PAVIA DATASET

Loss setting	MPSNR↑	MSSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑
$L_1 + 0.2L_{\mathrm{SAM}} + 0.1L_{\mathrm{grad}}$	29.2935	0.8110	5.5211	5.8585	0.0356	0.8917
$L_1 + 0.3L_{\mathrm{SAM}} + 0.1L_{\mathrm{grad}}$	29.3314	0.8124	5.4947	5.8324	0.0354	0.8926
$L_1 + 0.4L_{\mathrm{SAM}} + 0.1L_{\mathrm{grad}}$	29.3289	0.8120	5.4822	5.8374	0.0354	0.8925
$L_1 + 0.5L_{\mathrm{SAM}} + 0.1L_{\mathrm{grad}}$	29.3211	0.8124	5.4954	5.8382	0.0354	0.8926
$L_1 + 0.6L_{\mathrm{SAM}} + 0.1L_{\mathrm{grad}}$	29.3384	0.8131	5.4597	5.8257	0.0354	0.8929
$L_1 + 0.4L_{\mathrm{SAM}}$	29.3292	0.8127	5.4961	5.8329	0.0354	0.8927
$L_1 + 0.4L_{\mathrm{SAM}} + 0.2L_{\mathrm{grad}}$	29.3239	0.8128	5.5001	5.8363	0.0355	0.8924
$L_1 + 0.4L_{\mathrm{SAM}} + 0.3L_{\mathrm{grad}}$	29.3229	0.8132	5.5196	5.8390	0.0355	0.8923

Direct HR-space unfolding in Experiment II achieves lower MPSNR and MSSIM than the PST-DUNet, indicating that performing the main iterative refinement directly in the HR space may increase the reconstruction difficulty. Experiment III further removes the MR-space unfolding refinement and learns a direct LR–HR mapping without STR, which leads to a larger degradation in both spatial fidelity and spectral consistency. These results suggest that placing the main learned unfolding process in the MR space provides a more effective intermediate reconstruction domain between the LR observation and the HR target.

2) Contribution of STR: Comparing Experiment I with Experiment IV evaluates the role of STR. Removing STR increases SAM from 5.4822 to 5.6444 and RMSE from 0.0354 to 0.0357 while also reducing MPSNR and MSSIM. This indicates that injecting the topology-aggregated spectral representation into spatial feature refinement is beneficial for both spectral consistency and spatial fidelity. Experiment V replaces the STR gating mechanism with simple concatenation fusion. Its performance is clearly inferior to the PST-DUNet, showing that topology information should be used as conditional modulation rather than only appended as additional features.

3) Contribution of DSTI: Experiments VII and VIII further analyze the role of DSTI. In Experiment VII, the dynamic interband aggregation is removed, and only self-loop node updating is retained. In Experiment VIII, a fixed full graph is used instead of input-dependent dynamic topology inference. Both variants degrade the reconstruction performance, especially on SAM and ERGAS. These results indicate that the adaptive interband dependency inferred from the current reconstruction state provides more useful spectral coordination than self-only updating or a fixed graph structure.

4) Contribution of Temperature Scaling: Experiment VI fixes the temperature coefficient to $\tau = 1$. Compared with PST-DUNet, this setting leads to lower MPSNR and higher SAM, indicating that the learnable temperature coefficient helps adaptively adjust the concentration of the topology weights and improves the reliability of interband aggregation.

D. Sensitivity Analysis

1) Sensitivity to Loss Weights: To evaluate the influence of the loss weights, we conduct a sensitivity analysis on λ_{sam} and λ_{grad}. Based on the default setting $\lambda_{sam} = 0.4$ and $\lambda_{grad} = 0.1$, we vary one weight at a time while keeping the other fixed. The results on Pavia Center ×4 and Chikusei ×4 are reported in Tables V and VI.

Overall, PST-DUNet is not highly sensitive to moderate changes in the loss weights. Increasing λ_{sam} generally helps improve SAM, while the overall reconstruction metrics remain stable. For λ_{grad}, excessively large weights do not bring consistent improvement, suggesting that the gradient term should be used as a complementary structural constraint. Therefore, we use $\lambda_{sam} = 0.4$ and $\lambda_{grad} = 0.1$ as the default setting for all datasets and scale factors as the same as the common setting from the other methods.

2) Sensitivity to the Number of Unfolding Steps: We further analyze the influence of the MR unfolding step number K on the Chikusei ×4 dataset. As shown in Table VII, increasing K from one to three gradually improves the reconstruction performance. The setting $K = 3$ achieves the best overall balance in terms of MPSNR, MSSIM, SAM, ERGAS, RMSE, and CC. When K is further increased to 4, the parameter number continues to grow, but the performance no longer improves. This indicates that the benefit of additional unfolding steps becomes saturated around $K = 3$. Therefore, we set $K = 3$ in all main experiments.

E. Model Complexity Analysis

We further compare the model complexity of PST-DUNet with representative baseline methods in terms of parameter number and FLOPs. The FLOPs are computed under the Pavia Center ×4 setting. The results are shown in Table VIII.

PST-DUNet has 26.9188 M parameters and 108.9169 G FLOPs. Compared with lightweight methods such as EUNet and SNLSR, PST-DUNet introduces higher computational cost. This additional cost mainly comes from the MR-space learned unfolding process, DSTI, and STR-based topology-conditioned feature modulation. Compared with very large-parameter models such as RCAN, PST-DUNet uses fewer parameters, but its FLOPs are higher than several CNN-based baselines because the main refinement is performed in the MR space.

Instead, PST-DUNet aims to find a new reconstruction logic for HSI-SR than the common design in recent years. Future work may further reduce the computational cost through

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TABLE VI
SENSITIVITY ANALYSIS OF LOSS WEIGHTS ON ×4 CHIKUSEI DATASET

Loss setting	MPSNR↑	MSSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑
$L_1 + 0.2L_{\text{SAM}} + 0.1L_{\text{grad}}$	42.8818	0.9595	1.7109	4.0861	0.0093	0.9587
$L_1 + 0.3L_{\text{SAM}} + 0.1L_{\text{grad}}$	42.8973	0.9595	1.7074	4.0621	0.0093	0.9588
$L_1 + 0.4L_{\text{SAM}} + 0.1L_{\text{grad}}$	42.8840	0.9595	1.7005	4.0591	0.0093	0.9586
$L_1 + 0.5L_{\text{SAM}} + 0.1L_{\text{grad}}$	42.8892	0.9594	1.7030	4.0610	0.0093	0.9587
$L_1 + 0.6L_{\text{SAM}} + 0.1L_{\text{grad}}$	42.9169	0.9596	1.6947	4.0489	0.0093	0.9589
$L_1 + 0.4L_{\text{SAM}}$	42.8900	0.9593	1.7067	4.0682	0.0093	0.9585
$L_1 + 0.4L_{\text{SAM}} + 0.2L_{\text{grad}}$	42.9046	0.9594	1.7119	4.0442	0.0093	0.9588
$L_1 + 0.4L_{\text{SAM}} + 0.3L_{\text{grad}}$	42.8841	0.9594	1.7109	4.0794	0.0093	0.9586

TABLE VII
SENSITIVITY ANALYSIS OF THE UNFOLDING STEP NUMBER K ON ×4 CHIKUSEI DATASET

Setting	MPSNR↑	MSSIM↑	SAM↓	ERGAS↓	RMSE↓	CC↑	Param. (M)
$K = 1$	42.7985	0.9587	1.7233	4.1001	0.0094	0.9578	16.56
$K = 2$	42.8139	0.9588	1.7219	4.0955	0.0094	0.9580	22.04
$K = 3$	42.8840	0.9595	1.7005	4.0591	0.0093	0.9586	26.91
$K = 4$	42.8520	0.9591	1.7216	4.0792	0.0093	0.9583	33.01

TABLE VIII
MODEL COMPLEXITY COMPARISON ON ×4 PAVIA DATASET

Method	Param. (M)	FLOPs (G)
RCAN	82.9993	23.6695
RDN	5.6644	6.3590
GDRNN	1.6497	93.7301
EUNet	0.7799	15.2354
CTUNet	9.0721	83.0546
SSPSR	13.3847	145.5761
SNLSR	1.5948	3.3588
CST	10.9868	19.8642
PST-DUNet	26.9188	108.9169

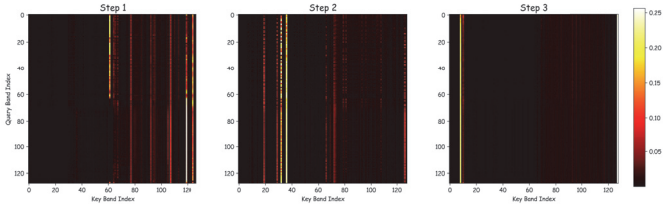


Fig. 7. Visualization of the spectral affinity matrices across different unfolding stages.

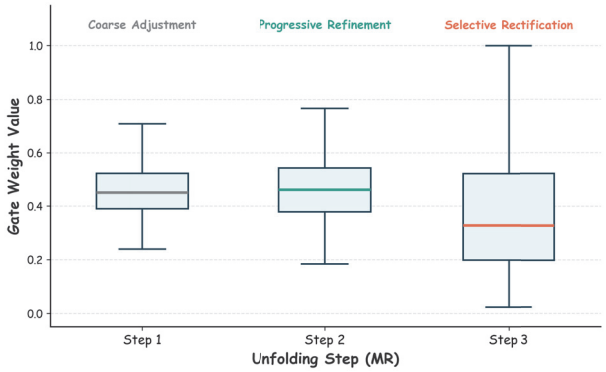


Fig. 8. Evolution of the feature rectification mechanism.

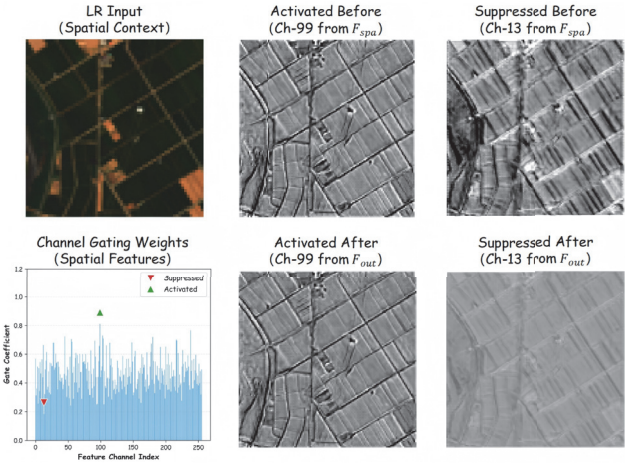


Fig. 9. Visualization of the feature rectification effect within the STR.

parameter sharing across unfolding steps, lightweight spectral topology inference, and low-rank adjacency approximation.

F. Visualization Analysis

1) Visualization of Dynamic Spectral Topology: To further understand the internal behavior of PST-DUNet, we visualize the evolution of the dynamic spectral adjacency matrices inferred by DSTI and the channelwise gates generated by STR, as illustrated in Figs. 7–9. These visualizations are not used as independent proof of performance improvement,

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but as qualitative evidence for the proposed reconstruction mechanism.

The affinity matrices show that DSTI does not use a fixed spectral graph for all inputs or all unfolding steps. Instead, the interband dependency patterns vary with the input and the reconstruction stage, which is consistent with the design motivation of reconstruction-state-conditioned spectral topology inference. Meanwhile, the gate visualization shows that STR produces different channelwise modulation responses according to the topology-aggregated spectral representation. This indicates that the inferred interband dependency is not only visualized as an isolated relation matrix, but is further transformed into feature modulation for spatial refinement.

Together with the quantitative ablation results, these visualizations support the role of DSTI and STR in coordinating spatial detail recovery with spectral consistency preservation.

G. Discussion

Although PST-DUNet achieves promising performance under the adopted nonblind bicubic degradation setting, several issues deserve further investigation.

First, PST-DUNet is not designed as a blind degradation HSI-SR method. The DC module explicitly computes the consistency residual between the projected MR estimate and the observed LR HSI. Therefore, when the real degradation process deviates substantially from the bicubic degradation setting used for paired data construction, the reliability of the observation-consistency correction may be affected. Although the learnable projection and back-projection operators provide a certain degree of flexibility, they are learned under the training degradation distribution and should not be interpreted as blind degradation estimators.

Second, the dynamic spectral adjacency matrix in DSTI introduces a quadratic bandwise cost with respect to the number of spectral bands. This cost is acceptable for the benchmark datasets used in this article, but may become more significant for sensors with substantially larger spectral dimensions. Sparse, low-rank, or top-k adjacency approximation may be explored in future work.

Third, the current experiments follow dataset-specific training and testing. Generalization across sensors with different spectral response functions or wavelength samplings may require additional designs, such as wavelength-aware embeddings, spectral response alignment, or degradation-adaptive training.

V. Conclusion

In this article, we propose PST-DUNet, a PST-DUNet for HSI-SR. PST-DUNet follows a fixed LR–MR–HR reconstruction path, where the MR space is used as an intermediate reconstruction domain to reduce the difficulty of direct large-scale upsampling. In the MR stage, observation-consistency correction is combined with learned prior refinement, enabling the LR observation to guide the progressive recovery of spatial details. To improve spectral consistency during reconstruction, we design DSTI to infer reconstruction-state-conditioned interband dependencies from the current image-domain estimate. The topology-aggregated spectral representation is then transformed by STR into spatially shared channelwise modulation for spatial feature refinement, which helps coordinate spatial detail recovery with spectral consistency preservation. Experiments on the evaluated public HSI datasets under the adopted nonblind bicubic degradation setting demonstrate that PST-DUNet achieves superior or competitive performance compared with representative HSI-SR methods. Ablation studies further verify the effectiveness of the progressive MR-space refinement, DSTI, and STR. In future work, we will investigate degradation-adaptive data consistency and lightweight spectral topology inference to further improve the robustness and efficiency of HSI-SR.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] J. Yang, T. Lin, X. Chen, and L. Xiao, "Multiple deep proximal learning for hyperspectral-multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525814.

[3] F. A. Kruse and S. L. Perry, "Regional mineral mapping by extending hyperspectral signatures using multispectral data," in *Proc. IEEE Aerosp. Conf.*, Mar. 2007, pp. 1–14.

[4] D. Krezhova, S. Maneva, N. Petrov, I. Moskova, and K. Krezhov, "Detection of environmental changes using hyperspectral remote sensing," *AIP Conf. Proc.*, vol. 1722, no. 1, 2016, Art. no. 270003.

[5] Q. Xie et al., "Leaf area index estimation using vegetation indices derived from airborne hyperspectral images in winter wheat," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 8, pp. 3586–3594, Aug. 2014.

[6] J. Yang, Y.-Q. Zhao, J. C.-W. Chan, and S. G. Kong, "Coupled sparse denoising and unmixing with low-rank constraint for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1818–1833, Mar. 2016.

[7] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," 2023, *arXiv:2311.07113*.

[8] X. Sun, L. Zhuang, L. Gao, H. Gao, X. Sun, and B. Zhang, "Information entropy estimation based on point-set topology for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5523415.

[9] P. Ren et al., "Oil spill drift prediction enhanced by correcting numerically forecasted sea surface dynamic fields with adversarial temporal convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4701018.

[10] W. Zhang, H. Wang, P. Ren, and W. Zhang, "Underwater image color correction via color channel transfer," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[11] G. A. Shaw and H. K. Burke, "Spectral imaging for remote sensing," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 3–28, 2003.

[12] R. Wu, T. Yang, L. Sun, Z. Zhang, S. Li, and L. Zhang, "SeeSR: Towards semantics-aware real-world image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25456–25467.

[13] J.-N. Su, M. Gan, G.-Y. Chen, W. Guo, and C. L. P. Chen, "High-similarity-pass attention for single image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 610–624, 2024.

[14] C. Tian, Y. Zhang, W. Zuo, C.-W. Lin, D. Zhang, and Y. Yuan, "A heterogeneous group CNN for image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6507–6519, May 2024.

[15] B. Li et al., "SeD: Semantic-aware discriminator for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25784–25795.

[16] J. Kim, J. Oh, and K. M. Lee, "Beyond image super-resolution for image recognition with task-driven perceptual loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2651–2661.

[17] L. Wang, J. Li, Y. Wang, Q. Hu, and Y. Guo, "Learning coupled dictionaries from unpaired data for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25712–25721.

[18] Y. Long, X. Wang, M. Xu, S. Zhang, S. Jiang, and S. Jia, "Dual self-attention Swin transformer for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512012.

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[19] Y. Xu et al., "Hyperspectral image super-resolution with ConvLSTM skip-connections," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519016.
[20] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601514.
[21] S. Jiang, N. Li, M. Xu, S. Zhang, and S. Jia, "SQformer: Spectral-query transformer for hyperspectral image arbitrary-scale super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532315.
[22] Q. Li, Q. Wang, and X. Li, "Mixed 2D/3D convolutional network for hyperspectral image super-resolution," *Remote Sens.*, vol. 12, no. 10, p. 1660, May 2020.
[23] Z. Shi, C. Chen, Z. Xiong, D. Liu, Z.-J. Zha, and F. Wu, "Deep residual attention network for spectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2018, pp. 214–229.
[24] X. Wang, J. Ma, and J. Jiang, "Hyperspectral image super-resolution via recurrent feedback embedding and spatial–spectral consistency regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503113.
[25] H. Huang, A. G. Christodoulou, and W. Sun, "Super-resolution hyperspectral imaging with unknown blurring by low-rank and group-sparse modeling," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2014, pp. 2155–2159.
[26] Y. Wang, X. Chen, Z. Han, and S. He, "Hyperspectral image super-resolution via nonlocal low-rank tensor approximation and total variation regularization," *Remote Sens.*, vol. 9, no. 12, p. 1286, Dec. 2017.
[27] T. Akgun, Y. Altunbasak, and R. M. Mersereau, "Super-resolution reconstruction of hyperspectral images," *IEEE Trans. Image Process.*, vol. 14, no. 11, pp. 1860–1875, Nov. 2005.
[28] H. Huang, J. Yu, and W. Sun, "Super-resolution mapping via multi-dictionary based sparse representation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 3523–3527.
[29] H. Irmak, G. B. Akar, S. E. Yuksel, and H. Aytaylan, "Super-resolution reconstruction of hyperspectral images via an improved MAP-based approach," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 7244–7247.
[30] H. Irmak, G. B. Akar, and S. E. Yuksel, "A MAP-based approach for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2942–2951, Jun. 2018.
[31] S. Mei, X. Yuan, J. Ji, Y. Zhang, S. Wan, and Q. Du, "Hyperspectral image spatial super-resolution via 3D full convolutional neural network," *Remote Sens.*, vol. 9, no. 11, p. 1139, Nov. 2017.
[32] J. Li, R. Cui, B. Li, R. Song, Y. Li, and Q. Du, "Hyperspectral image super-resolution with 1D–2D attentional convolutional neural network," *Remote Sens.*, vol. 11, no. 23, p. 2859, Dec. 2019.
[33] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.
[34] Y. Li, L. Zhang, C. Dingl, W. Wei, and Y. Zhang, "Single hyperspectral image super-resolution with grouped deep recursive residual network," in *Proc. IEEE 4th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2018, pp. 1–4.
[35] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.
[36] Y. Yuan, X. Zheng, and X. Lu, "Hyperspectral image superresolution by transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1963–1974, May 2017.
[37] Q. Hu, X. Wang, J. Jiang, X.-P. Zhang, and J. Ma, "Exploring the spectral prior for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5260–5272, 2024.
[38] Y. Liu, J. Hu, X. Kang, J. Luo, and S. Fan, "Interactformer: Interactive transformer and CNN for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531715.
[39] S. Chen, L. Zhang, and L. Zhang, "Cross-scope spatial–spectral information aggregation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 33, pp. 5878–5891, 2024.
[40] Y. Xu et al., "MambaHSISR: Mamba hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511216.
[41] D. Liu et al., "An efficient unfolding network with disentangled spatial–spectral representation for hyperspectral image super-resolution," *Inf. Fusion*, vol. 94, pp. 92–111, Jun. 2023.
[42] K. Wang, X. Liao, J. Li, D. Meng, and Y. Wang, "Hyperspectral image super-resolution via knowledge-driven deep unrolling and transformer embedded convolutional recurrent neural network," *IEEE Trans. Image Process.*, vol. 32, pp. 4581–4594, 2023.
[43] Y. Tang et al., "A CNN-transformer embedded unfolding network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525016.
[44] Y. Zhang, K. Li, K. Li, L. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.
[45] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

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